Theory of Computation

Revlew of Mathematical Theory

1-7 Let us discuss each one by one -

l) Negation

This proposition is also called NOT proposition. It is denoted by I or ~ or having horizontal bar on letter.

If S is a proposition then S defines NOr s. The truth table for this connective is

	1s
T	F-
F	1

Conjunction (AND)

If A and B are two propositions then conjunction of A and B is denoted by A^B. The truth table for this connective is

A	B	A^B
T	T	T
T	F	F
F	T	F
F	.F	F

lii) Disjunction OR)

If A and B are two propositions then disjunction of A and B is denoted by AvB. The truth table for this connective is

A	B	AvB
T	T	T
T	F	T
F	T	T
F	F	F

iv) Implication

If A and B are two propositions then if A then B is a proposition. It can be denoted byA-BorA = B.



Figure 1

Roviow of Mathematical Theory

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Theory of Computation

The truth table for this connective is

AB B A

T T T

F F T

T T F

T F F

For example Find truth value for following proposition.

If an apple is not green then it must be red".

Let. A denotes "An apple is not green" and B denotes It must be red The statement A is false and B is true. Hence the proposition is true.

v) If and only if

If A and B are two propositions then A if and only f B. It can be denoted as AB

The truth table will be

A	B	AB
T	T	T
T	F	F
F	T	F

A	B	AB
F	F	T

For example: I will go to the school if and only if it is not raining Here statement A is I will go to the school". The statement B is It is not raining. Thus the resultant proposition is false.

1.5 Relations

Relaonship is a major aspect between two objects, even this is true in our real life One ohfect can be related with the other object by a some relation. Then those two objects forma pair based on this certain relationship.

Definition The relation R is a colection for the set S which represents the pair of elements,

For example (a b is in R We can represent their relation as a R b. The first component of each pair is chosen from a set called domain and second component of each pair is chosen from a set called range.



Figure 1

Review of Mathomatical Theory

Theory of Computation

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Properties of Relations

A relation R on set S is

1. Reflexive if iRi for all i in S.
2. Irreflexive if iRi is false for all i in S.
3. Transitive if iRj and jRk imply iRk
4. Symmetric if iRj implies jRi.
5. Asymmetric if iRj implies that jRi is false.

Every asymmetric relation must be irreflexive.

For example: ifA = a. b then

reflexive relation R can be = a, ab, b

irreflexive relation R can be =a, b)

transitive relation R can be = a. b). b, aa. a

symmetric relation R can be=|a b. ba

asymmetric relation R can be =a, b

Equivalent Relation

A relation is said to be equivalence relation if it is reflexive, symmetric and transitive, over some set S.

Suppose R is a set of relations and S is the set of elements. For example S is the set of lines in a plane and R is the relation of lines intersecting to each other.

A relation is said to be antisymmetric if a, be R and b, ae R then a =a.

Example 1.5: Determine whether R is equivalence relation or not where

A = 0121R = 0010.11)2, 221

Solution The R is reflexive because 0, 0)1, 1).2, 2E R.

Where as R is not symmetric because 0, 1) is not in R whereas (1, 0) is in R. Hence the R is not a equivalence relation.

Closures of Relations

Sometimes when the relation R is given it may not be reflexive or transitive By adding some pairs we make the relation as reflexive or transitive. For example Letab) b, c. aab, b